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theoria poiesis praxis

DEPARTAMENTO DE ELECTRÓNICA, TELECOMUNICAÇÕES E INFORMÁTICA
LICENCIATURA EM ENG. DE COMPUTADORES E INFORMÁTICA

REDES DE COMUNICAÇÃO I

LABORATORY GUIDE NO. 4

NAT/PAT, DHCP, SUBNETTING AND IPV6 BASICS

Objectives

- Study of the NAT/PAT mechanisms.
- Study of DHCP.
- Subnetting
- Study of IPv6

Duration

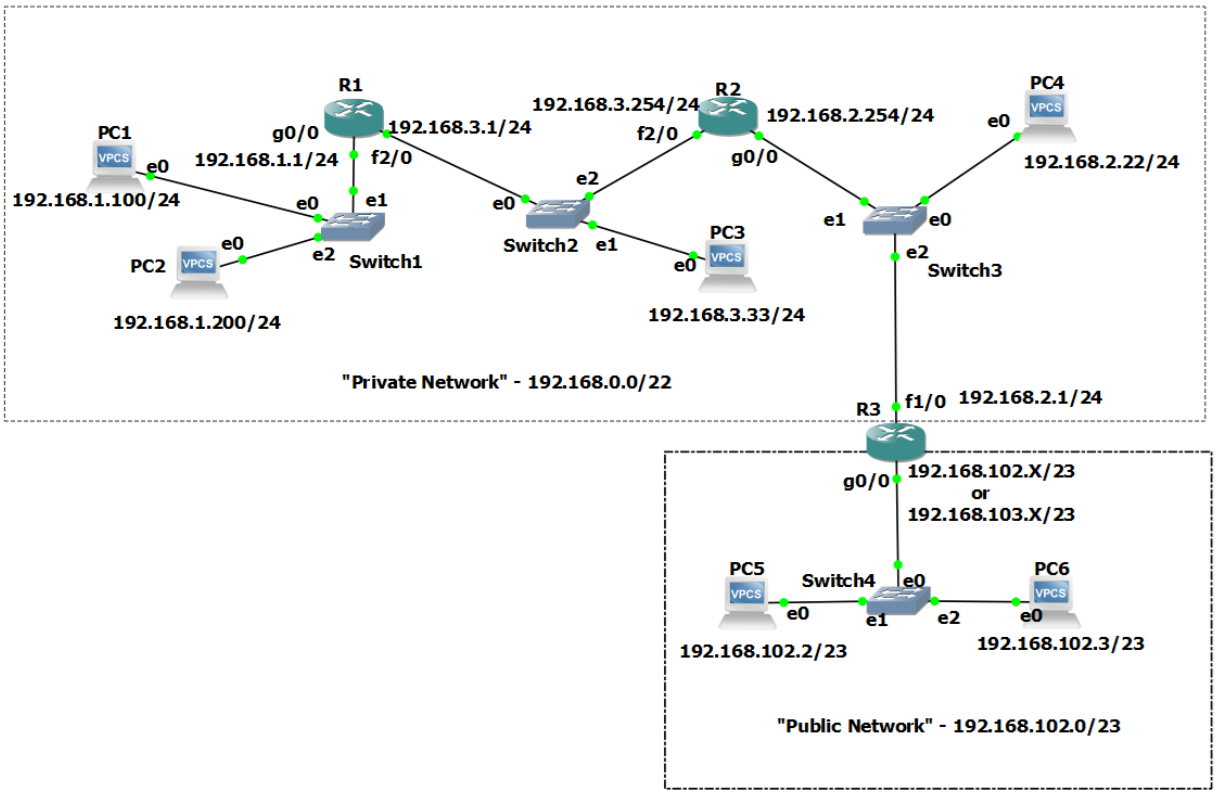
3 weeks

Part 1 – NAT/PAT

1. Dynamic NAT

1.1. Base Network configuration

Assemble and configure (using the GNS3 and VPCS hosts) the network depicted in the following figure which represents a small company network (this network is based on the network from Guide 2).
The company decided to configure IP private addressing using the network 192.168.0.0/22 and NAT mechanism (without PAT) to manage all Internet accesses.



a) Create default routes on R1 and R2 towards the "Public Network"

R1:
R1# **configure terminal** // conf t
R1(config)# **ip route 0.0.0.0 0.0.0.0 192.168.3.254**

R2:
R2# **configure terminal** // conf t
R2(config)# **ip route 0.0.0.0 0.0.0.0 192.168.2.1**

b) Configure R3 network interfaces (please refer to the annex to get the g0/0 IP address corresponding to your ID –"X" value) and routing for the internal networks.

R3:
R3# configure terminal // conf t
R3(config)# int g0/0
R3(config-if)# ip address 192.168.102/3.X 255.255.254.0 // Refer to the annex
R3(config-if)# no shut
R3(config)# int f1/0
R3(config-if)# ip address 192.168.2.1 255.255.255.0
R3(config-if)# no shut
R3(config-if)#exit
R3(config)#ip route 192.168.0.0 255.255.252.0 192.168.2.254 // R2 is the GW to the inside

c) Configure VPCs 5 and 6 with the corresponding IP addresses (do not configure gateway).

1.2. Dynamic NAT configuration

In order to define a pool of global addresses to be allocated by the dynamic NAT process, issue the following command on Router 3 (refer to the annex for the value of "Y"):

R3:
R3# **configure terminal** // conf t
R3(config)# ***ip nat pool MYNATPOOL 192.168.102/3.Y 192.168.102/3.Y netmask 255.255.255.0***

This command defines a pool with a single public address (192.168.102/3.Y).

The name **MYNATPOOL** is the name of the address pool. The first **192.168.102/3.Y** in the command is the first IP address in the pool, and the second **192.168.102/3.Y** is the last IP address in the pool (this command creates a pool that contains only a single address, since the first and the last value have the same IP address).

Next, configure a standard **access list** to define which internal source addresses can be translated. Since any users on the private network are being translated, use the following command:

R3(config)# ***access-list 2 permit 192.168.0.0 0.0.3.255***

This command defines that the networks 192.168.0.0/24, 192.168.1.0/24, 192.168.2.0/24 and 192.168.3.0/24 will be allowed (permitted) on the NAT process.

Finally, the establish the dynamic NAT translation, it is necessary to link the access list to the name of the NAT pool, as follows:

R3(config)# ***ip nat inside source list 2 pool MYNATPOOL***

Note that "source list 2" refers to the "access-list 2" created before, and "pool MYNATPOOL" refers to the "nat pool" created before (be careful to guarantee that the names are exactly the same)

Finally, specify **an interface on the Router to be used by inside network hosts** requiring address translation:

R3(config)# **interface f1/0**
R3(config-if)# **ip nat inside**

Also, specify **an interface to be used as the outside NAT interface** as follows:

R3(config)# **interface g0/0**
R3(config-if)# **ip nat outside**

NOTE: If you are using different router interfaces, please adapt the commands to match the interfaces you are using.

1.2.1. Start a packet capture on the public network (between R3 g0/0 and Switch4) and another on the private network (between R3 f1/0 and Switch3). At PC1 execute a ping to PC5 (192.168.102.2), and on PC2 execute a ping also to PC5 (192.168.102.2). Verify (on the router) the active NAT translations and NAT activity statistics, use the commands

R3# **show ip nat translations**
R3# **show ip nat statistics**

>> Which packets had the source IP addresses translated? Explain the obtained results.

1.2.2. Execute on the router the command to clear the NAT translation table:

R3# **clear ip nat translation ***

and execute again a ping at PC2 to **192.168.102.3**

>> Explain the observed results.

1.2.3. Change NAT timeout to 60 seconds and clear the NAT translations table:

R3(config)# ***ip nat translation timeout 60***
R3# **clear ip nat translation ***

At PC1 execute a ping to **192.168.102.2**, and immediately after, at PC2 execute repeatedly a ping to **192.168.102.2**. How much time does it take to obtain connectivity between PC2 and host **192.168.102.2**?

>> Explain the observed results.

1.2.4. Restore NAT timeout value to 86400 seconds (24 hours):

R3(config)# **ip nat translation timeout 86400**

1.3. Dynamic NAT/PAT

1.3.1. The most powerful feature of NAT is **address overloading, or port address translation (PAT)**. Overloading allows multiple inside addresses to map to a single global address. With PAT, the NAT router keeps track of the different conversations by mapping TCP and UDP port numbers. After defining the pool of global addresses to be allocated by the dynamic NAT process and configuring the standard access list that defines which internal source addresses can be translated, configure address overloading on Router with the following command:

R3(config)# **ip nat inside source list 2 pool MYNATPOOL overload**

Note: You may have to reset the active NAT translations with **clear ip nat translation ***

Repeat experience 1.2.1.

>>Which are the advantages of using NAT and PAT mechanisms?

1.3.2. From PC1 (and PC2) try to establish UDP and TCP connections (ports 80 and 22) to the host 192.168.102.2:

```
PC> ping 192.168.102.2 -2 -p 80          ! UDP port 80
PC> ping 192.168.102.2 -2 -p 22          ! UDP port 22
PC> ping 192.168.102.2 -3 -p 80          ! TCP port 80
PC> ping 192.168.102.2 -3 -p 22          ! TCP port 22
```

Note: The option -p must have a space after (before the port number).

>> Verify (on the router) the active NAT translations and NAT activity statistics. Explain the obtained results.

1.4. Static NAT/PAT Translations

1.4.1. Try to ping the private network machines (PC1, 2, 3 or 4 or router interfaces) from PC5 or PC6.

>> Explain the results.

1.4.1. Suppose that now you have another public IP address available (address W on the annex). Configure the router in order to allow the PC5 or 6 to access PC1.

A static translation between the inside local address of a host and one of the inside global addresses can be created using the following commands:

R3(config)# **ip nat inside source static 192.168.1.100 192.168.102/3.W**

From PC5, ping PC1's static public address (**192.168.102/3.W**)

PC5> **ping 192.168.102/3.W**

>> Analyze the captured packets on the private network and explain the obtained results.

>> Discuss a scenario where static NAT/PAT is required.

Part 2 - DHCP

2. DHCP

2.1. DHCP server on a Cisco Router.

2.1.1. Configure Router 1 (R1) as DHCP server for the private network. Assume that you want to dynamically assign addresses from the range 192.168.1.101 to 192.168.1.199

```
R1(config)# service dhcp
R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.100
R1(config)# ip dhcp excluded-address 192.168.1.200 192.168.1.254
R1(config)# ip dhcp pool 1
R1(dhcp-config)# network 192.168.1.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.1.1
```

Use the following commands to verify the configuration and status of the DHCP server:

```
R1# show ip dhcp pool
R1# show ip dhcp server statistics
R1# show ip dhcp binding
```

2.2. Start a capture on Router 1's G0/0 interface.

Configure PC1 to acquire the IPv4 address dynamically:

```
PC1> ip dhcp
```

Configure PC1 to renew the IPv4 address dynamically:

```
PC1> ip dhcp -r
```

Configure PC1 to release the IPv4 address dynamically:

```
PC1> ip dhcp -x
```

Configure PC1 to acquire again the IPv4 address dynamically:

```
PC1> ip dhcp
```

Note: you may need to reset PC1 IP address. (remove and save configuration without IP)

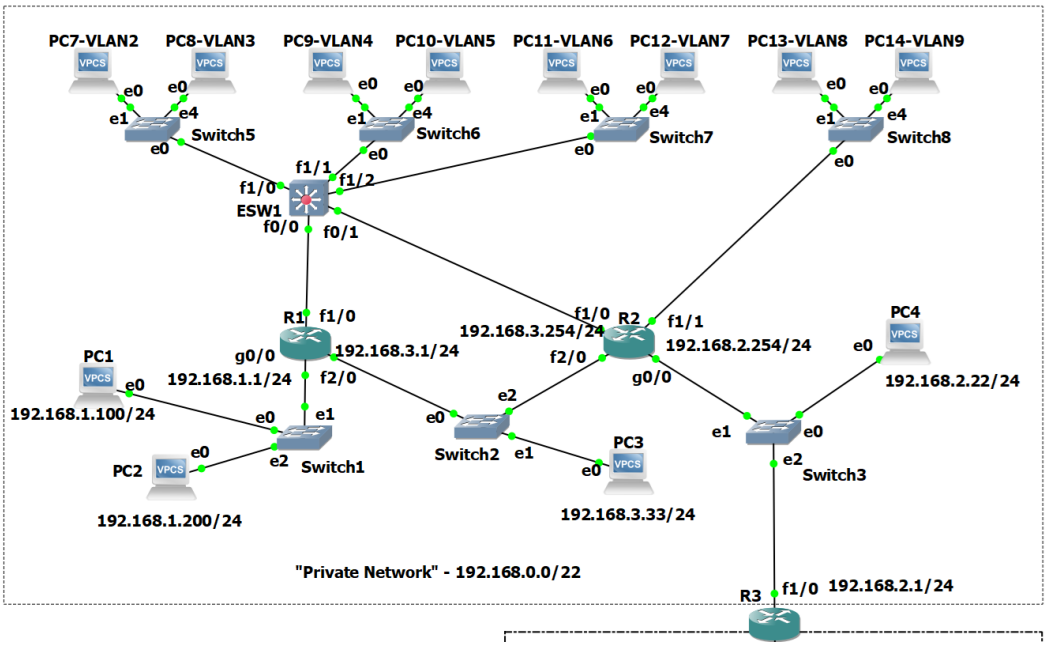
>> In each step, analyze the exchanged DHCP packets and the contents of the DHCP Bindings at Router 1.

At this point, you may configure DHCP Pools on R1 for the remaining networks on the private part. On R2, interface g0/0, you will need to add the command "**ip helper-address 192.168.3.1**" and provide the general **service dhcp** command.

Part 3 – Subnetting

2.2.1. Subnetting

Continue using the network from the previous section, add an ESW and 4 switches. Configure the network topology as follows:



Use the network 172.25.X.0/22 to provide addresses for all the VLANs above, according to the following requirements:

VLAN2 – at least 500 addresses + router/ESW/VLAN interface

VLAN3 – at least 16 addresses + router/ESW/VLAN interface

VLAN4 – at least 61 addresses + router/ESW/VLAN interface

VLAN5 – at least 125 addresses + router/ESW/VLAN interface

VLAN6 – at least 70 addresses + router/ESW/VLAN interface

VLAN7 – at least 30 addresses + router/ESW/VLAN interface

VLAN8 – at least 25 addresses + router/ESW/VLAN interface

VLAN9 – at least 5 addresses + router/ESW/VLAN interface

The router inter-connections (R1–ESW1 and R2-ESW1) must use /30 subnets from the same address space.

>> Divide the address space to meet the requirements. Present it on a table. Show how you did, using a diagram.

	Network	Mask	Broadcast	Gateway Addr.	Available Address Space
VLAN2	172.25.0.0	23	172.25.1.255	172.25.0.1	-
VLAN3	172.25.3.128	/27	172.25.3.159	172.25.3.129	-
VLAN4	172.25.3.0	/26	172.25.3.63	172.25.3.1	-
VLAN5	172.25.2.0	25	172.25.2.127	172.25.2.1	-
VLAN6	172.25.2.128	25	172.25.2.255	172.25.2.129	-
VLAN7	172.25.3.64	/26	172.25.3.127	172.25.3.65	-
VLAN8	172.25.3.192	/27	172.25.3.223	172.25.3.193	-
VLAN9	172.25.3.224	/29	172.25.3.231	172.25.3.225	-

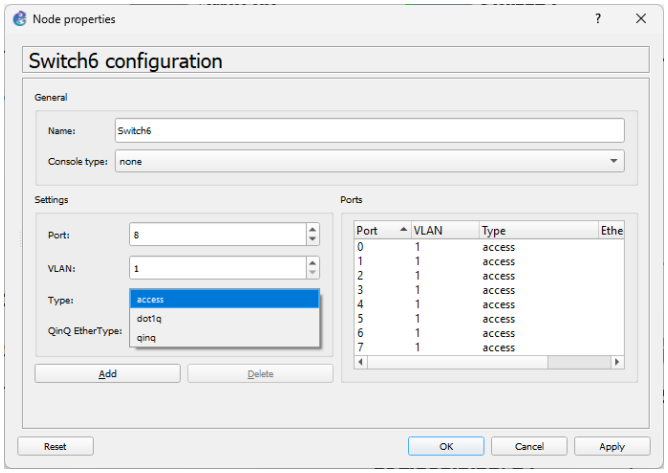
Interconnections		
NETWORK / MASK	**ESW1**	**R1/R2**
172.25.3.232 /30	172.25.3.234	172.25.3.233 (R1)
172.25.3.236 /30	172.25.3.238	172.25.3.237 (R2)

2.2.2. Configure the ESW and R2 with the specified VLANs

Refer to the VLANs guide to the specific commands. Do not forget to configure the ESW to perform "ipv4 routing" and "service dhcp".

2.2.3. Configure the switch ports to according to this scenario

Some ports must be dot1Q (trunk)



2.2.4. **Configure the interconnections**

>> Verify connectivity between the routers

2.2.5. Configure routing between the routers to allow full connectivity on all the network. Use a default route whenever is possible.

>> Present the routing tables of R1, R2, R3 and ESW.

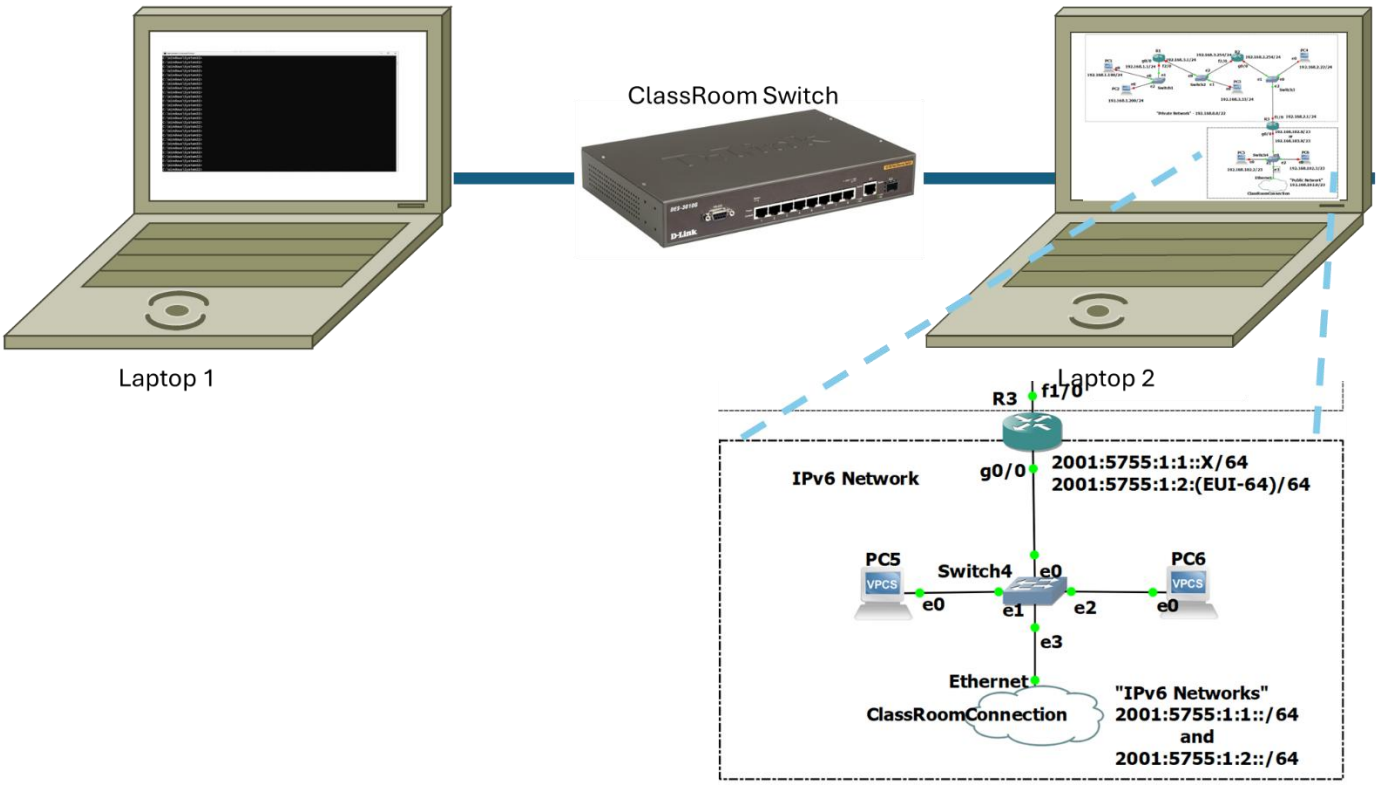
2.2.6. Configure DHCP pools to all the VLANs on R1. On R2 and on ESW, use the "ip helper-address" command on the VLAN interfaces, pointing to R1

Part 4 – Ipv6

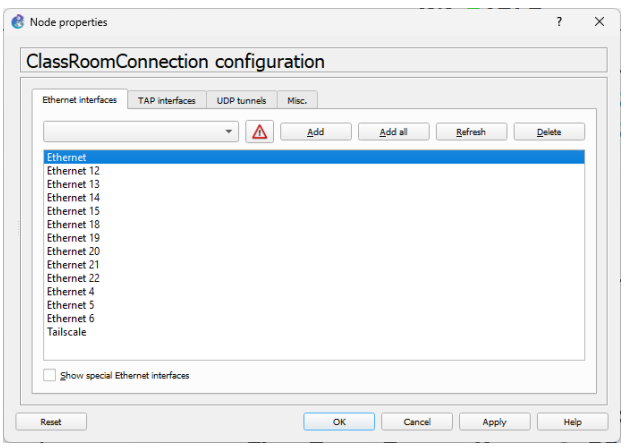
3. IPv6 Basic Mechanisms

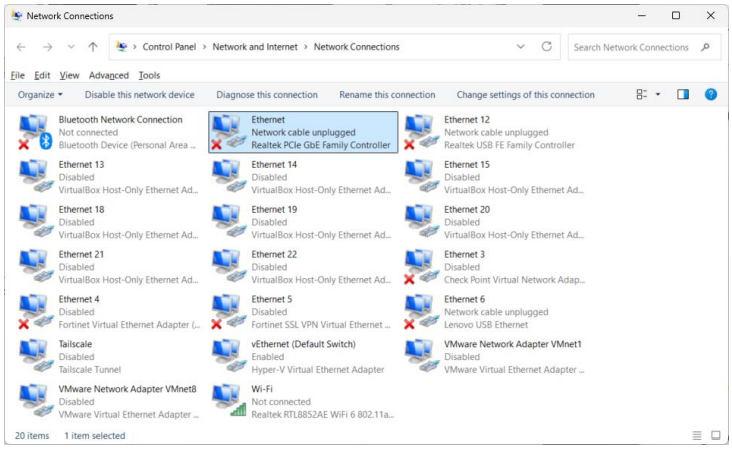
3.1.1. Continue using the same network and add a "cloud" as shown on the picture below (You may use the classroom PC as "Laptop 1"). Connect the Cloud to one of the switch ports and to the Ethernet network adapter of your laptop (if your laptop does not have a built-in Ethernet adapter, use a USB-to-Ethernet adapter).

(Note: you may cross-configure this scenario: each Laptop will simultaneously act as Laptop 1 and Laptop 2)



The connection on the cloud must be done to the correct interface, corresponding to the Ethernet adapter. If you "right click" on the Cloud, you may choose "Configure" to check if the right adapter is present. On the PC ("view network connections" on Windows), you may check the name of the Ethernet adapter.





3.1.2. Start a capture in the link between the Cloud and the switch inside GNS3. Turn off and back on the Laptop1 Ethernet interface. Both laptops must have the IPv6 protocol enabled on the Ethernet interfaces.

>>Stop the capture and analyze the IPv6 packets. Do you see any IPv6 packets? If yes, which? If not, do you know why?

Re-start the capture in the link between the Cloud and the Switch. Power on R3 (if it is powered down) and configure its interface to network A (g0/0).

```
R3(config)# ipv6 unicast-routing          //enable IPv6 routing on this router
R3(config)# interface g0/0
R3(config-if)# ipv6 enable                //enable IPv6 routing on this interface (g0/0)
R3(config-if)# no shutdown
```

Verify router's interfaces names and configuration:

```
R3# show ipv6 interface
R3# show ipv6 interface brief
```

Restart PC's Ethernet interface and verify its interface information:

Windows: ipconfig /all

```
Linux:  sudo ifconfig eth0 down     (sudo ip link set eth0 down)
        sudo ifconfig eth0 up       (sudo ip link set eth0 up)
        ifconfig eth0               (sudo ip address)
```

* it may not be “eth0” – Check with “sudo ifconfig”

Stop the capture and analyze the IPv6 packets and equipment’s information. Use the following commands to verify interfaces' IPv6 addressing and verify router's IPv6 routing table:

```
R3#show ipv6 interface brief
R3#show ipv6 route
```

>> Register the output from the previous commands

3.1.3. Re-start a capture in the link between the PC and the Switch. Configure Router's interface with a manually defined IPv6 global address from network 2001:5755:1:1::/64.

```
R3(config)# interface g0/0
R3(config-if)# ipv6 address 2001:5755:1:1::X/64   //refer to the annex for the values of "X")
```

>> Verify Laptop 1 Ethernet interface information.

Stop the capture and analyze the IPv6 packets.

>> Verify the Router's interfaces IPv6 addresses and the router's IPv6 routing table.

>> Explain the process by which the Laptop 1 obtained the IPv6 addresses.

3.1.4. Re-start a capture in the link between the Cloud and the Switch.

Configure Router's interface in order to add a second IPv6, this time with a EUI-64 based IPv6 global address from network 2001:5755:1:2::/64.

```
R3(config)# interface g0/0
R3(config-if)# ipv6 address 2001:5755:1:2::/64 eui-64
```

>> Verify Laptop 1 Ethernet interface information.

Stop the capture and analyze the IPv6 packets.

>>Verify the Router's interfaces IPv6 addressing and the router's IPv6 routing table.

>> Explain the process by which the Router completed the last 64 bits of its IPv6 addresses.

>> Discuss a possible disadvantage of using the standard EUI-64 at routers' interfaces.

>> Does the process, by which the PC obtained the IPv6 addresses, change by using the EUI-64 standard at the Router?

3.1.5. Re-start a capture in the link between the Cloud and the Switch.

At the laptop, using the command ping6 perform a ping to:

a) Router's R3 Link-Local address

Linux: you need to define the output interface with option "-I eth0" or "-I enp4s0".

Windows: you need to specify the source address (ping -S srcaddr destaddr).

b) Router's Global address from network 2001:5755:1:1::/64.

c) Router's Global address from network 2001:5755:1:2::/64.

Stop the capture and analyze the IPv6/ICMPv6 packets.

>> Explain the physical addresses resolution process in IPv6.

At this point, go to PC5 and PC6 and perform the command "ip auto" and check if they both get a global IPv6 address (due to a limitation on the VPC, unlike the Laptops, the VPCs can only have one IPv6 address).

ANNEX A – List of IP Addresses to be used by each student.

Mec Num	Name	"Public GW Address" (192.168.102/3.X)	"NAT Pool Address" (192.168.102/3.Y)	"NAT Fixed Address" (192.168.102/3.W)
72476	MARIA LUISA RODRIGUES DA SILVEIRA LEITE	192.168.102.4	192.168.102.5	192.168.102.6
85085	DIOGO ALEXANDRE DIAS	192.168.102.8	192.168.102.9	192.168.102.10
96661	NICOLE MONTEIRO RAKOV	192.168.102.12	192.168.102.13	192.168.102.14
98610	PEDRO MIGUEL MARTINHO GUERRA	192.168.102.16	192.168.102.17	192.168.102.18
103751	FRANCISCO MIGUEL FERREIRA ALMEIDA	192.168.102.20	192.168.102.21	192.168.102.22
107987	GABRIEL ESTÊVÃO SAL MONTEIRO	192.168.102.24	192.168.102.25	192.168.102.26
108115	MARTA BEATRIZ BESSA RIBEIRO PEREIRA PEDROSA	192.168.102.28	192.168.102.29	192.168.102.30
108154	TOMÁS CORREIA DE SÁ DOS SANTOS ALVES	192.168.102.32	192.168.102.33	192.168.102.34
108770	LUÍS DANIEL ESCAROUPA PEREIRA	192.168.102.36	192.168.102.37	192.168.102.38
110509	RUI PEREIRA DE MELO SILVA DE ALBUQUERQUE	192.168.102.40	192.168.102.41	192.168.102.42
113417	RITA CARDOSO SEQUEIRA	192.168.102.44	192.168.102.45	192.168.102.46
114525	DANIEL COSTEIRA NARCISO	192.168.102.48	192.168.102.49	192.168.102.50
114982	MARCOS MATOS KOUFALIOTIS	192.168.102.52	192.168.102.53	192.168.102.54
115739	OGBONNA AMOBI SAMUEL	192.168.102.56	192.168.102.57	192.168.102.58
115876	GUILHERME DA COSTA GRAÇA	192.168.102.60	192.168.102.61	192.168.102.62
117127	FELIPE NERES GOMES E SILVA	192.168.102.64	192.168.102.65	192.168.102.66
117596	GUILHERME CORRÊA DE VASCONCELOS MAIA	192.168.102.68	192.168.102.69	192.168.102.70
118781	PEDRO MIGUEL DA SILVA CORADO	192.168.102.72	192.168.102.73	192.168.102.74
119372	GUILHERME TAVARES PEREIRA	192.168.102.76	192.168.102.77	192.168.102.78
119844	JOANA CASTRO E SILVA	192.168.102.76	192.168.102.77	192.168.102.78
119928	MADALENA MESQUITA CEREJEIRA AMARO DIOGO	192.168.102.80	192.168.102.81	192.168.102.82
120139	JOÃO PEDRO GÉNIO DE MAGALHÃES	192.168.102.84	192.168.102.85	192.168.102.86
120300	ÂNGELO MANUEL CAUCHIE FIGUEIREDO	192.168.102.88	192.168.102.89	192.168.102.90
120353	VIACHESLAV SEREDA	192.168.102.92	192.168.102.93	192.168.102.94
121728	JOÃO MOISÉS KUTAYA	192.168.102.96	192.168.102.97	192.168.102.98
122892	LAURA MORAES TAKIGAMI GARCIA	192.168.102.100	192.168.102.101	192.168.102.102
123430	SANJAY THARU	192.168.102.104	192.168.102.105	192.168.102.106
123436	SILLAS VALADARES ARAÚJO	192.168.102.108	192.168.102.109	192.168.102.110
124686	RODRIGO MIGUEL CRUZ ANDRADE	192.168.102.112	192.168.102.113	192.168.102.114
124693	DAVIDE DOS SANTOS SILVA	192.168.102.116	192.168.102.117	192.168.102.118
124765	ANDRÉ HILÁRIO BRAZ	192.168.102.120	192.168.102.121	192.168.102.122
124779	DANIEL FERREIRA CARVALHO	192.168.102.124	192.168.102.125	192.168.102.126
124855	RODRIGO FERREIRA DE MORAIS	192.168.102.128	192.168.102.129	192.168.102.130
125085	JOÃO MARIA ANASTÁCIO AZINHEIRA	192.168.102.132	192.168.102.133	192.168.102.134
125160	VICENTE AMORIM SILVA	192.168.102.136	192.168.102.137	192.168.102.138
125191	JOÃO DANIEL SOARES DE MELO	192.168.102.140	192.168.102.141	192.168.102.142
125293	RICARDO FRANCISCO ASSEMBLEIA FRANCISCO	192.168.102.144	192.168.102.145	192.168.102.146
125329	PEDRO ESTEVES FONTES	192.168.102.148	192.168.102.149	192.168.102.150
125432	VASCO QUEIRÓS FERREIRA	192.168.102.152	192.168.102.153	192.168.102.154
125478	VICTOR MIGUEL LOUREIRO VAZ DA COSTA MORAIS	192.168.102.156	192.168.102.157	192.168.102.158
125738	INES VEIGAS CARDOSO	192.168.102.160	192.168.102.161	192.168.102.162
125753	JOÃO PEDRO LOPES CEREJO	192.168.102.164	192.168.102.165	192.168.102.166
125761	AFONSO PINTO CORREIA COELHO	192.168.102.168	192.168.102.169	192.168.102.170
125828	LOURENÇO GOMES SECO	192.168.102.172	192.168.102.173	192.168.102.174
125939	AFONSO GIL PEREIRA SAMAGAIO	192.168.102.176	192.168.102.177	192.168.102.178
125977	MIGUEL JOSÉ CORDEIRO LEAL	192.168.102.180	192.168.102.181	192.168.102.182
125986	EDUARDO CONSTANTINO VAZ ALVES	192.168.102.184	192.168.102.185	192.168.102.186
125996	TIAGO FILIPE ALMEIDA TRUTA	192.168.102.188	192.168.102.189	192.168.102.190
126023	PEDRO MIGUEL FARO RODRIGUES MARTELO	192.168.102.192	192.168.102.193	192.168.102.194

Mec Num	Name	"Public GW Address" (192.168.102/3.X)	"NAT Pool Address" (192.168.102/3.Y)	"NAT Fixed Address" (192.168.102/3.W)
126132	MARTIM JESUS PEREIRA DE ALMEIDA	192.168.102.196	192.168.102.197	192.168.102.198
126154	MARCO ANTÓNIO CRUZ FLOR	192.168.102.200	192.168.102.201	192.168.102.202
126193	EDGAR SIMÃO NOVO PASCOAL	192.168.102.204	192.168.102.205	192.168.102.206
126253	PEDRO SANTOS TAVARES	192.168.102.208	192.168.102.209	192.168.102.210
126265	SAMUEL JOSÉ MÓNICA CARVALHAIS	192.168.102.212	192.168.102.213	192.168.102.214
127378	AFONSO MIGUEL LOUREIRO MELO	192.168.102.216	192.168.102.217	192.168.102.218
127408	ANTÓNIO PEDRO DE ANDRADE ESCADA	192.168.102.220	192.168.102.221	192.168.102.222
127535	ALEXANDRE MOTSNY	192.168.102.224	192.168.102.225	192.168.102.226
127548	GONÇALO MEDEIROS MOURA	192.168.102.228	192.168.102.229	192.168.102.230
128120	RAFAEL DOS SANTOS FERREIRA	192.168.102.232	192.168.102.233	192.168.102.234
129499	HAILIN WU	192.168.102.236	192.168.102.237	192.168.102.238
129508	ARTHUR SUZART SPANÓ	192.168.102.240	192.168.102.241	192.168.102.242
129520	LORENZO STANISLAW LAVALIER LIMA	192.168.102.244	192.168.102.245	192.168.102.246
131821	AFONSO FIGUEIREDO MARQUES OLIVEIRA	192.168.102.248	192.168.102.249	192.168.102.250
131898	HENRIQUE GUTMANN GALA	192.168.102.252	192.168.102.253	192.168.102.254
131970	FILIPE DOS SANTOS GONÇALVES	192.168.103.4	192.168.103.5	192.168.103.6
131973	JOÃO BENTO CARREIRA	192.168.103.8	192.168.103.9	192.168.103.10
132006	DAVID DE ALMEIDA DIAS	192.168.103.12	192.168.103.13	192.168.103.14
132052	LUCAS JACINTO PEREIRA	192.168.103.16	192.168.103.17	192.168.103.18
132063	TOMÁS SOBRAL SILVA	192.168.103.20	192.168.103.21	192.168.103.22
132101	PEDRO GUILHERME ELOY CARVALHO	192.168.103.24	192.168.103.25	192.168.103.26
132163	TOMÁS FILIPE FONSECA MARQUES	192.168.103.28	192.168.103.29	192.168.103.30
132171	LUCAS HENRIQUE ALECRIM SILVA	192.168.103.32	192.168.103.33	192.168.103.34
132205	DANIEL KAMOLOV	192.168.103.36	192.168.103.37	192.168.103.38
132279	JOANA MARTINS FERREIRA	192.168.103.40	192.168.103.41	192.168.103.42
132346	JOÃO RODRIGO DA COSTA NUNES	192.168.103.44	192.168.103.45	192.168.103.46
132398	MARIA JOÃO DA SILVA FERREIRA	192.168.103.48	192.168.103.49	192.168.103.50
132425	JOSÉ ANDRÉ MARTINS DA ROCHA	192.168.103.52	192.168.103.53	192.168.103.54
132440	GONCALO BORDONHOS BARBOSA	192.168.103.56	192.168.103.57	192.168.103.58
132443	TIAGO GIL REBELO DE JESUS	192.168.103.60	192.168.103.61	192.168.103.62
132536	TOMÁS ANDRÉ GENRINHO PATO	192.168.103.64	192.168.103.65	192.168.103.66
132556	GUILHERME MANUEL LOURENÇO MANTA	192.168.103.68	192.168.103.69	192.168.103.70
132593	GONÇALO SANTOS	192.168.103.72	192.168.103.73	192.168.103.74
132655	IURY DA COSTA FIGUEREDO	192.168.103.76	192.168.103.77	192.168.103.78
132663	AFONSO RODRIGO SANTOS	192.168.103.80	192.168.103.81	192.168.103.82
132677	ANTÓNIO OLIVEIRA LIMA	192.168.103.84	192.168.103.85	192.168.103.86
132678	LUCA ZAGNOLI	192.168.103.88	192.168.103.89	192.168.103.90
132694	RAÚL LUÍS BASULO RIBEIRO	192.168.103.92	192.168.103.93	192.168.103.94
132716	GONÇALO SANTOS REIS	192.168.103.96	192.168.103.97	192.168.103.98
132807	GABRIELA SOFIA SILVA ADREGO FONSECA	192.168.103.100	192.168.103.101	192.168.103.102
132883	GONÇALO SOARES OLIVEIRA	192.168.103.104	192.168.103.105	192.168.103.106
132909	ANA TERESA PEREIRA TAVARES	192.168.103.108	192.168.103.109	192.168.103.110
133010	AFONSO MARTIM FONSECA BALTAZAR SILVA	192.168.103.112	192.168.103.113	192.168.103.114
133031	MARCO ANTONIO BORGES OLIVEIRA	192.168.103.116	192.168.103.117	192.168.103.118
133048	RODRIGO MIGUEL GOMES FIGUEIREDO	192.168.103.120	192.168.103.121	192.168.103.122
133140	RUI PEDRO ANDRADE CANOTILHO CORREIA	192.168.103.124	192.168.103.125	192.168.103.126
133185	RODRIGO MIGUEL CORREIA DOS SANTOS	192.168.103.128	192.168.103.129	192.168.103.130
133256	PEDRO PINTO MARIA	192.168.103.132	192.168.103.133	192.168.103.134
133287	ALEXANDRU ZAVGORODNII	192.168.103.136	192.168.103.137	192.168.103.138
133291	FRANCISCO SANTOS BRANCO	192.168.103.140	192.168.103.141	192.168.103.142
133304	MIGUEL SILVA CRUZ BICHÃO	192.168.103.144	192.168.103.145	192.168.103.146

Mec Num	Name	"Public GW Address" (192.168.102/3.X)				"NAT Pool Address" (192.168.102/3.Y)				"NAT Fixed Address" (192.168.102/3.W)			
133443	GUSTAVO DAVID MOREIRA DE SOUSA	192.168.	103	.	148	192.168.	103	.	149	192.168.	103	.	150
133454	SANTIAGO DUQUE GONZÁLEZ	192.168.	103	.	152	192.168.	103	.	153	192.168.	103	.	154
133486	LUCAS OLIVEIRA SANTOS	192.168.	103	.	156	192.168.	103	.	157	192.168.	103	.	158
134191	FRANCISCO CAMPANTE MARQUES FERREIRA	192.168.	103	.	160	192.168.	103	.	161	192.168.	103	.	162
134206	SALVADOR PINTO FERREIRA	192.168.	103	.	164	192.168.	103	.	165	192.168.	103	.	166
134274	BRUNO OLIVEIRA RODRIGUES	192.168.	103	.	168	192.168.	103	.	169	192.168.	103	.	170
134282	SIMÃO BORGES SARAIVA	192.168.	103	.	172	192.168.	103	.	173	192.168.	103	.	174
134283	FRANCISCO NUNES ESTEVES DE CAMPOS LOURDES	192.168.	103	.	176	192.168.	103	.	177	192.168.	103	.	178
134309	DINIS MIRA PEREIRA DE ABREU	192.168.	103	.	180	192.168.	103	.	181	192.168.	103	.	182
134317	ALEXANDRE NOGUEIRA MORAIS	192.168.	103	.	184	192.168.	103	.	185	192.168.	103	.	186
134352	JOÃO PEDRO GOMES MELO DE ALMEIDA	192.168.	103	.	188	192.168.	103	.	189	192.168.	103	.	190
134357	RODRIGO LEIRIA FERREIRA	192.168.	103	.	192	192.168.	103	.	193	192.168.	103	.	194
134427	GUILHERME MARCELO FERRAZ	192.168.	103	.	196	192.168.	103	.	197	192.168.	103	.	198
134549	LARA ALEXANDRA GARRIDO PRETO	192.168.	103	.	200	192.168.	103	.	201	192.168.	103	.	202
135021	SÉRGIO ANDRÉ DE SOUSA VALENTE	192.168.	103	.	204	192.168.	103	.	205	192.168.	103	.	206
135025	LUIS DAVI DA SILVA RIBEIRO	192.168.	103	.	208	192.168.	103	.	209	192.168.	103	.	210
137892	DANIEL STRUK	192.168.	103	.	212	192.168.	103	.	213	192.168.	103	.	214
141845	ANDRÉ CÉSAR FERREIRA	192.168.	103	.	216	192.168.	103	.	217	192.168.	103	.	218